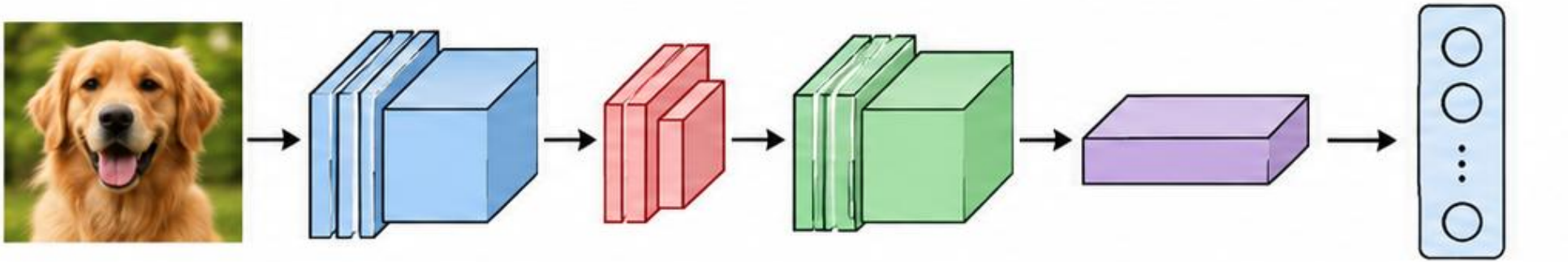
CNN pipeline for image classification

Conceptual architecture: local filters build a hierarchy of visual features.



Input image
224 × 224 × 3
(RGB)

Convolution + ReLU
learned local filters
(e.g., 3 × 3)

Pooling
spatial reduction
(e.g., 2 × 2, stride 2)

Convolution blocks
deeper feature maps
(e.g., 64 → 128 → 256)

Global average pool
compact descriptor
(e.g., 1 × 1 × 256)

Classifier
class probabilities
(e.g., 1000 classes)

How spatial evidence is handled

Shared kernels impose locality and translation-related structure; receptive fields grow with depth.
The diagram is schematic and does not imply a fixed number of layers, parameters, or measured accuracy.